Chiral quotients with small regular covers

Human Compression project

Hartley, Hubard and Leemans, *Two atlases of abstract chiral polytopes for small groups*, Ars Mathematica Contemporanea 5 (2012), 371–382, report 56 chiral polytopes whose minimal regular covers have at most 2000 flags, excluding covers with 1024 or 1536 flags. We verify the 56 listed examples using the literal companion data acquired on 15 October 2026. This note proves their existence and distinctness. Exhaustiveness among all covers is a separate obligation and is not asserted here. A distinct dual is counted separately, while the two enantiomorphic markings describe one unrooted polytope.

For each entry the retained data specify permutations $r_0, \dots, r_{d-1}$ generating W, and words generating a subgroup $N \leq W$. Put $H = \langle r_0 r_i : 1 \leq i < d \rangle$. The vertices of the flag graph are the right cosets Nw; the edge of colour i sends Nw to Nwr_i. These operations are unambiguous. We acquire the previously proved normalizer theorem: the group of automorphisms preserving every colour is $N_W(N)/N$. In particular, if $N_W(N) = H$ has index two and avoids every r_i, the graph has two automorphism orbits, and each adjacency exchanges them. To obtain a polytope one must also verify the incidence axioms; normalizer data alone do not imply them.

Lemma 1 (Reconstruction from a finite flag graph)**.** *Let a finite graph have d specified involutory adjacencies, none of which fixes a vertex. For each i, call the connected components using all colours except i its faces of rank i. Associate to each graph vertex its tuple of faces. Suppose the following finite checks hold: these tuples are distinct; incidence defined by occurrence in a tuple is transitive; its maximal chains are exactly the tuples; every section of rank two has exactly two intermediate faces; and the graph vertices containing any fixed chain are connected by the remaining colours. Then adjoining a least and a greatest face gives an abstract polytope with precisely the specified flag graph.*

Proof. Ranks orient incidence between distinct ranks. Transitivity, together with equality within a rank, gives a partial order. Each face occurs in a tuple, and each comparable pair does so by definition. Every chain extends in the finite order to a maximal chain, which by hypothesis is a full tuple. Thus all maximal chains have the required length. The section checks are exactly the diamond axiom, including sections incident with the least or greatest face. The connectedness hypothesis is strong flag connectivity. An edge of colour i preserves every other component, and its endpoints differ in the remaining face by injectivity of tuples. The diamond axiom makes this the unique adjacent flag of that colour. Hence the given graph is exactly the flag graph. □

Lemma 2 (Complete comparison of coloured graphs)**.** *Starting at a chosen vertex, traverse a connected graph in breadth-first order, considering colours in a fixed order and numbering each vertex when it is first encountered. The resulting numbered adjacency table is a complete invariant of the graph with its chosen root. Trying each possible image of a root therefore finds all colour-preserving isomorphisms. The least rooted table is a complete unrooted invariant.*

Proof. A rooted colour-preserving isomorphism must carry the first traversal to the second, step by step, so it preserves all assigned numbers and the resulting table. Conversely, equality of tables gives the explicit bijection between equally numbered vertices, preserving every edge. Connectedness makes an isomorphism unique once its root image is fixed. Taking every possible root proves both assertions. □

Theorem 1 (The listed examples)**.** *The retained 56 entries give 48 pairwise nonisomorphic chiral polytopes of rank three and eight of rank four. Each has its stated face numbers, regular monodromy cover and rotation-group SmallGroup identifier.*

Proof. The literal permutation groups have the source orders and SmallGroup identifiers. For every pair of subsets of generators the checker verifies the complete intersection property. It also checks involutions, string commutation and each adjacent product order. Thus these are the stated regular polytopes, using the retained string C-group correspondence.

The stated words generate subgroups of exactly the stated orders. In each case $N \leq H$, $[W : H] = 2$, $N_W(N) = H$, and $\mathrm{Core}_W(N) = 1$. The quotient H/N has the stated SmallGroup identifier, verified on that actual quotient. The core test makes the flag action faithful, so its monodromy group is the literal group W. The normalizer theorem identifies the full automorphism group and its two flag orbits.

For all 19,712 cosets the checker constructs the complete adjacency table. It computes every face component, verifies incidence transitivity, enumerates every maximal chain and checks all diamonds. For each subset of ranks it groups flags by their common faces and compares the resulting sets with components in the complementary colours. Thus every chain residue is connected, and the reconstruction lemma applies. The computed face counts agree with every source entry.

Finally the second lemma checks every possible root image directly from the graphs, without using the group normalizer calculation. For each graph the images of a fixed root under its automorphisms are exactly the even vertices in its bipartition. Their number equals the stated rotation-group order. The 56 least adjacency tables are distinct. This proves chirality, full automorphism orders and the stated equivalence convention independently at the graph level. □

The rank-four rows. The four entries below and their reversals are the eight rank-four examples. We write (n, j) for SmallGroup(n, j); a group order alone is never used to identify a group.

Type	Regular cover group	Rotation group	$\lvert N\rvert$
$\{3, 3, 8\}$	$(768, 1086052)$	$(192, 1494)$	2
$\{3, 6, 9\}$	$(972, 108)$	$(162, 19)$	3
$\{3, 6, 18\}$	$(1944, 949)$	$(324, 77)$	3
$\{6, 6, 9\}$	$(1944, 949)$	$(324, 77)$	3

Certificates, attribution and remaining scope. The pages are parsed as literal integers, disjoint cycles and group words; no downloaded program is executed. Exact source bytes, generated calls and terminal receipts are retained. GAP's finite permutation-group, normalizer, coset-action and SmallGroup identification algorithms remain explicit software premises. The incidence and isomorphism checks use only the complete finite adjacency tables and ordinary integer operations. The rotation relators displayed on the pages are also satisfied, but this does not prove that every displayed abstract presentation has no additional elements. Presentation equality is not claimed by that test.

The source authors retain credit for the examples and classification. The normalizer argument, incidence reconstruction and deterministic rooted comparison supply a uniform verification. No new open-question solution or compression measurement is asserted. The earlier integral proof for the order-588 example is acquired separately. Present website bytes are not asserted to be identical to the 2012 website. A complete reconstruction of the source still needs exclusion of all other regular covers, and its separate almost-simple atlas.